


Mathematical Foundations of Generative AI

- Introduction to probabilistic ML
- variational Methods (latent variable Models)
 - Mixture Models (GMM)
 - Variational Auto-Encoder
 - Diffusion Models (DDPMs)
- Adversarial family
 - Generative Adversarial Networks (GANs)
- Auto-Regressive Models
 - Transformer-based LMs
- State-space Models (SSM)
- Flow-based Models

Probabilistic Machine Lear.

- Model the uncertainty / non-measurable abstracts

Eg: a) Classify a body of text as spam or not-a-spam.

b) Given an image, identify a particular person/object.

Model these systems via repeated observations.

- probabilistic approach.

Introduction to probability

A "random" experiment

- tossing a coin, rolling a die
- taking a picture, writing an email.

outputs a set of outcomes.

$$\Omega = \{ H, F \} : \text{coin toss}$$

$$\Omega = \{ 1, 2, 3, \dots, 6 \} : \text{die roll}$$

$$\Omega = \{ \text{pictures of all humans} \}$$

$$\Omega = \{ \text{texts in the internet} \}$$

Ω is sample space : set of all possible outcomes of a random experiment.

Define probability measure on Ω .

The idea of a measure,

$$\mu : \text{subsets} \rightarrow \mathbb{R}^+$$

Consider $\mathbb{R}$, & $A, B \subset \mathbb{R}$,

$$\text{Eg } A = [2, 4]$$

$$B = [6, 16]$$

Compare the "size" of A & B ,

we can define a size measure

$$A \subset \mathbb{R}, \quad A = [a, b],$$

$$\mu(A) = |b - a| \quad : \text{leb. measure}$$

Question : How to define a measure that quantifies the idea

of "uncertainty" associated with outcomes?

$\mathcal{F}$: set of subsets of Ω .

↳ Event space.

$$A \in \mathcal{F} \subset \Omega$$

Define a measure on the elements of $\mathcal{F}$, as follows:

$$P : \mathcal{F} \rightarrow [0, 1] : \text{probability measure}$$

properties of P :

a) $P(A) \geq 0$, $\forall A \in \mathcal{F}$

b) $P(\Omega) = 1$, $P(\emptyset) = 0$

c) $A, B \in \mathcal{F}$, $A \cap B = \emptyset$, then

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$$

Given there is a random experiment, there is a triplet $(\Omega, \mathcal{F}, \mathbb{P})$: probability triplet

Sample space Event space probability measure.

In practice, one would not have access to prob. triplet.

Eg: taking an X-ray image as random experiment.

We get to measure a surrogate of the elements of Ω (measurements)

We define another function on Ω ,
to describe the surrogate.

$$X : \Omega \rightarrow \mathbb{R}^d$$

Random Variable : function
from sample space onto real-
numbers.

Once a function (R.v.) is
defined on Ω , the prob. triplet
also gets "transformed".

$$(\Omega, \mathcal{F}, \mathbb{P}) \xrightarrow{X} (\mathbb{R}^d, \mathcal{B}, \mathbb{P}_X)$$

$$X : \Omega \rightarrow \mathbb{R}^d$$

$\mathcal{F}$ subset of Ω , $\mathcal{B}$ subset of
 $\mathbb{R}^d$.

$$P_x(x) = P\left(A: X^{-1}\left[(-\infty, x]\right)\right)$$

$$X: \Omega \rightarrow \mathbb{R}$$

$$X^{-1}: \mathbb{R} \rightarrow \Omega$$

$A \in \mathcal{F}$, this set gets mapped to a subset in $\mathbb{R}$ under X .

suppose this set $A \in \mathcal{F}$, gets mapped to $(-\infty, x]$ under X .

$\downarrow$
 subset of $\mathbb{R}$

then,

$$P_x(x) = P(A)$$

$$= P\left\{A: X^{-1}\left[(-\infty, x]\right)\right\}$$

= probability dist. function

prob. distribution function under a random variable X , evaluated at x $[P_X(x)]$ is the probability of the event A , that gets mapped to $(-\infty, x]$ under X .

$$P_X(x) = P\left(A : X \in (-\infty, x]\right)$$

$$\begin{array}{ccc} (\Omega, \mathcal{F}, \mathbb{P}) & \xrightarrow[\text{Rand. var}]{X} & (\mathbb{R}, \mathcal{B}, \mathbb{P}_X) \\ \downarrow & & \downarrow \end{array}$$

not accessible

abstract

measure &

work with